A deterministic emulator of the Veros ACC last-year state

28 October 2026

Summary

- Goal: predict the last-year temperature of the Veros ACC channel directly from the EKE-closure coefficients (c_k, c_ϵ), skipping the 100-year integration.
- Method: a small deterministic network $z \mapsto 20$ leading PCA coefficients of the training states, $z = (\log c_k, \log c_\epsilon)$ standardised, averaged over 5 initialisations. It fits on the run states in seconds on a CPU, with no GPU and no sampling.
- Interpolation (scattered points, a band of three rows, the 5×5 and 7×7 centre blocks, leave-one-out): RMSE 0.016–0.037 K. The nearest training run gives 0.038–0.118 K and the best classical emulator, a Gaussian process on one collapsed parameter s, 0.019–0.046 K.
- Extrapolation (top row, outer 51, outer 75): 0.059 / 0.068 / 0.123 K, against 0.079 / 0.075 / 0.125 K for the GP on s and 0.176 / 0.134 / 0.159 K for the nearest run. The warm corner sets the error: 0.15 K one row from the training set, 1.06 K three cells away.
- With only four training runs it loses to the GP on s (0.139 vs 0.120 K).

1 Data

- 100 Veros ACC runs on a 10×10 log-spaced grid, $c_k \in [0.0125, 0.8]$, $c_\epsilon \in [0.0875, 5.6]$ (ratio $2^{2/3}$); 100 years, 30-day output; interior grid $30 \times 42 \times 15$ (x, y, z) with a 930-cell ridge.
- Target: $y_r \in \mathbb{R}^{17970}$, the mean temperature of the last 12 outputs (360 days) of run r on the water cells. $S = 35$ psu everywhere, so it is dropped.
- The runs still drift, so the averaging window matters. Over the last 20 years, the RMS change from the first to the last year is 0.16 K, and the last-year mean differs from the 20-year mean by 0.08 K.
- The two parameters act through one combination, $s = \log c_k - 1.08 \log c_\epsilon$ (Figure 1). Below $s \approx -2$ (48 runs) the domain-mean T stays within 0.09 K; above, it rises by 1.1 K.

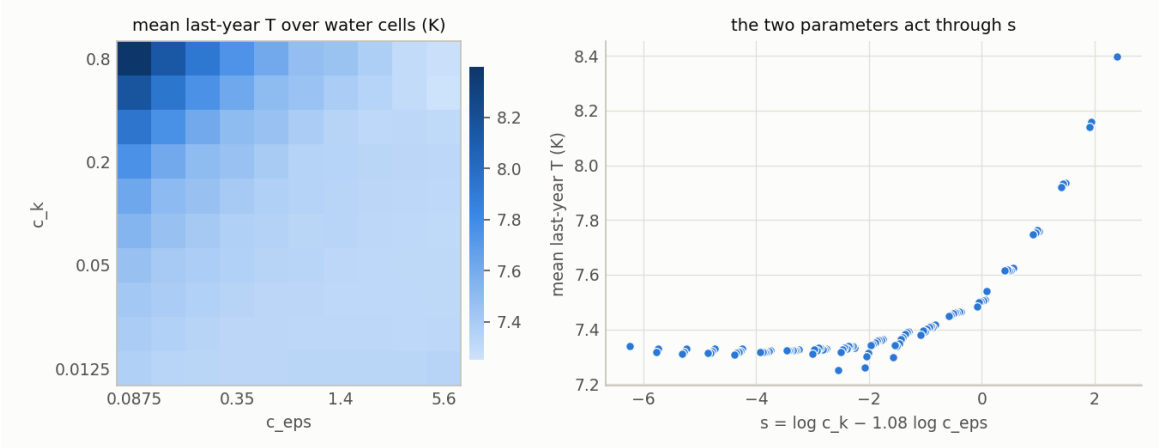


Figure 1: Mean last-year T over the water cells: over the parameter grid (left) and against s (right), the direction along which the first PCA coefficient of the states varies linearly.

2 Method

- Input z: $(\log c_k, \log c_\epsilon)$ standardised over the design.
- Output: a field in the span of the training states,

$$\hat{y}(z) = \mu + \sum_{k=1}^{K} a_k(z)\, v_k, \qquad K = 20, \tag{1}$$

  where μ and v_k are the mean and the leading principal directions of the training states.
- Network $a(z)$: a multilayer perceptron $2 \to 64 \to 64 \to K$ (scikit-learn, L-BFGS), with targets $a_k(z_r) = v_k^\top (y_r - \mu)$ divided by $\mathrm{std}(a_1)$. With orthonormal v_k, the loss is then the physical MSE of the projected field up to a constant:

$$\mathcal{L} = \frac{1}{n} \sum_{r=1}^{n} \frac{\|a(z_r) - V^\top (y_r - \mu)\|^2}{\mathrm{std}(a_1)^2} + \frac{\alpha}{n} \|W\|^2. \tag{2}$$

- The prediction is the mean of 5 networks that differ only in their initialisation.
- (activation, α) $\in \{\mathrm{ReLU}, \tanh\} \times \{10^{-4}, \ldots, 1\}$ is chosen by 5-fold cross-validation on the training runs of each split. The choice is ReLU with $\alpha = 10^{-4}$ to 10^{-2} everywhere except two splits: tanh on the 7×7 block, and $\alpha = 1$ with four training runs, where the cross-validation is not meaningful. The scores are flat for $\alpha \leq 10^{-2}$.
- Fitting on run means rather than on snapshots changes nothing: $\sum_t \|f(z_r) - x_{r,t}\|^2 = n_t \|f(z_r) - \bar{x}_r\|^2 + \mathrm{const}$.
- Baselines:
  - the training mean;
  - the nearest training run in z;
  - thin-plate RBF interpolation of every cell;
  - a GP per PCA coefficient on z (Matérn 5/2, one length scale per input, white noise);
  - the same GP on $s = w^\top z$, with w from a linear fit of a_1 on z.

3 Evaluation

- Eight hold-out splits of the grid, plus leave-one-out over the 100 runs:
  - interpolation: 10 scattered interior runs; the rows $c_k = 0.126$, 0.2, 0.3175; the centre 5×5 and 7×7 blocks;
  - extrapolation: the top row ($c_k = 0.8$); the 51 or 75 outer runs, training on the centre;
  - four training runs, one step in from each corner.
- Metric: RMSE over the water cells against the last-year mean, per held-out run, then the mean and median over the runs.
- Cost: a fit takes seconds on one core and a prediction microseconds. The whole evaluation takes 25 min on 8 cores, mostly the hyper-parameter search.

split	held out / train	network (median)	GP on s	GP on z	RBF	nearest run	mean state
scattered	10 / 90	**0.016** (0.013)	0.019	0.022	0.021	0.038	0.161
band (three rows)	30 / 70	**0.037** (0.032)	0.046	0.051	0.057	0.118	0.204
centre 5×5	25 / 75	**0.026** (0.023)	0.029	0.037	0.037	0.080	0.162
centre 7×7	49 / 51	**0.035** (0.032)	0.036	0.070	0.068	0.116	0.203
leave-one-out	100 / 99	**0.024** (0.017)	0.028	0.032	0.034	0.085	0.221
top row	10 / 90	**0.059** (0.033)	0.079	0.083	0.084	0.176	0.482
outer 51	51 / 49	**0.068** (0.028)	0.075	0.088	0.093	0.134	0.265
outer 75	75 / 25	**0.123** (0.057)	0.125	0.164	0.138	0.159	0.226
four runs	96 / 4	0.139 (0.127)	**0.120**	0.234	0.146	0.181	0.260

Table 1: RMSE (K) of the last-year temperature over the water cells, mean over the held-out runs; for the network also the median. Bold: lowest mean per split.

4 Findings

- **The network is best in 8 of 9 evaluations** (Table 1, Figure 2). In interpolation its error is half or less of the nearest run's.
- **Inside the grid it barely beats the GP on s.** On the 7×7 block it gets 0.035 vs 0.036 K and wins 26 runs of 49; in leave-one-out it wins 61 of 100. Since the parameters act almost only through s, a one-dimensional GP is a strong reference.
- **Its gain is at the grid edges.** On the top row it gets 0.059 vs 0.079 K, and at the corner run 0.15 vs 0.37 K. In leave-one-out, its largest errors are at the two high-c_k corners: 0.16 and 0.12 K (Figure 3).
- **The error sits in the upper** 500 m (Figures 4 and 5). In interpolation it is 0.02–0.05 K there and 0.01–0.02 K below 1000 m. At the centre of the 7×7 block, the predicted anomaly has a pattern correlation of 0.87 with the truth and 80 % of its amplitude; the error is largest at 300–500 m next to the southern boundary.
- **Large gaps defeat it.** On outer 75 (25 training runs) its median is 0.057 vs 0.038 K for the GP on s. With four runs it gets 0.139 vs 0.120 K, and in the flat regime the nearest run is better still (median 0.064 K).
- **The warm corner** ($c_k = 0.8$, $c_\epsilon = 0.0875$) is the hardest state (Figure 5).
  - One row from the training set (top row), the network reproduces its warming: correlation 0.99, amplitude 1.01, 0.15 K.
  - Three cells away (outer 75), it keeps a weak version of the pattern (correlation 0.56) with 30 % of the amplitude: a cold bias of −0.71 K and 1.06 K RMSE.
  - Every method is above 1 K there.
- **Part of that is the basis.** Projecting the held-out truth on the 20 training principal directions already leaves 0.010–0.017 K in interpolation and 0.031–0.079 K in extrapolation. On outer 75 that is 0.079 K of the network's 0.123 K.
- **The spread over initialisations is not an uncertainty.** It is 0.005–0.016 K in interpolation and 0.013–0.024 K in extrapolation, while the error grows from 0.016 to 0.139 K.

5 Next steps

- Inputs $(s, s_\perp)$ instead of $(\log c_k, \log c_\epsilon)$, to build in the collapse that makes the GP on s strong with few runs.
- For extrapolation, either a decoder not restricted to the span of the training states, or more runs near the warm corner.
- An uncertainty from resampling the training runs rather than the initialisations.
- Fixed hyper-parameters (ReLU, $\alpha = 10^{-3}$): the cross-validation scores are flat, and the evaluation would run about 50 times faster.

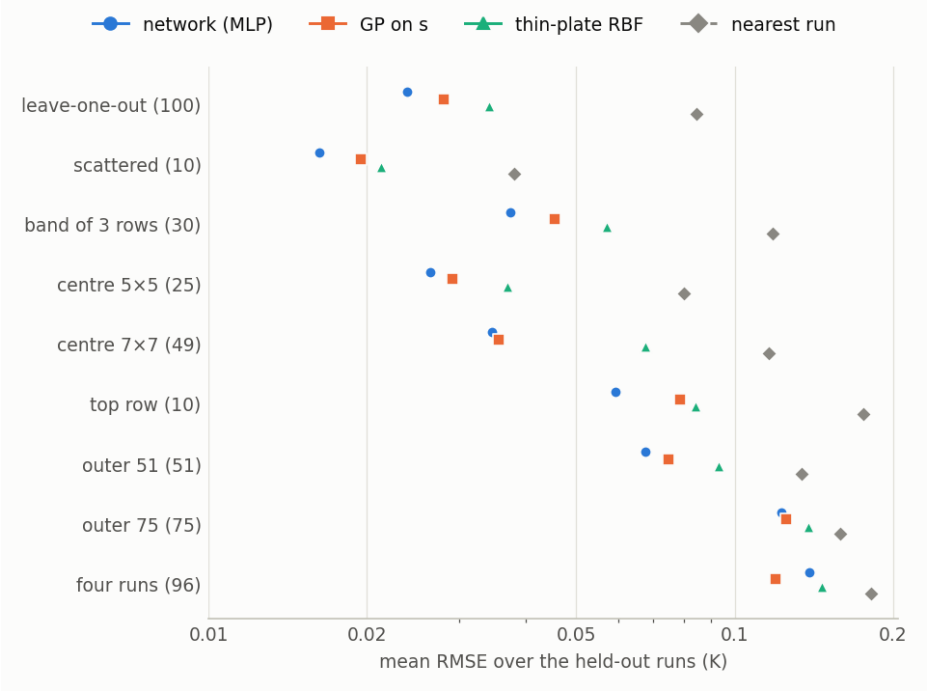


Figure 2: Mean RMSE over the held-out runs per split (log scale), for the network and the three best baselines; held-out run counts in brackets.

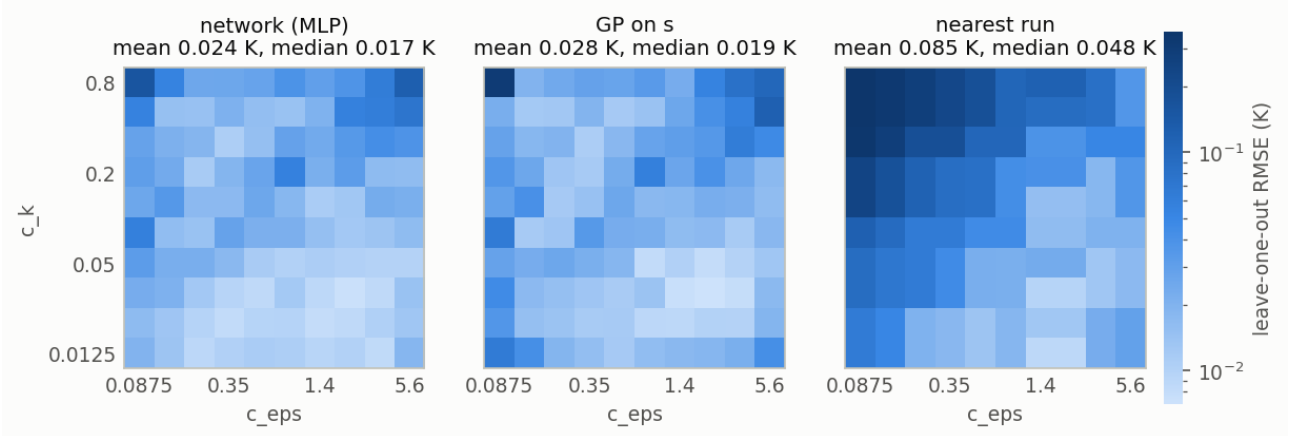


Figure 3: Leave-one-out RMSE of every run over the parameter grid (each run predicted from the 99 others).

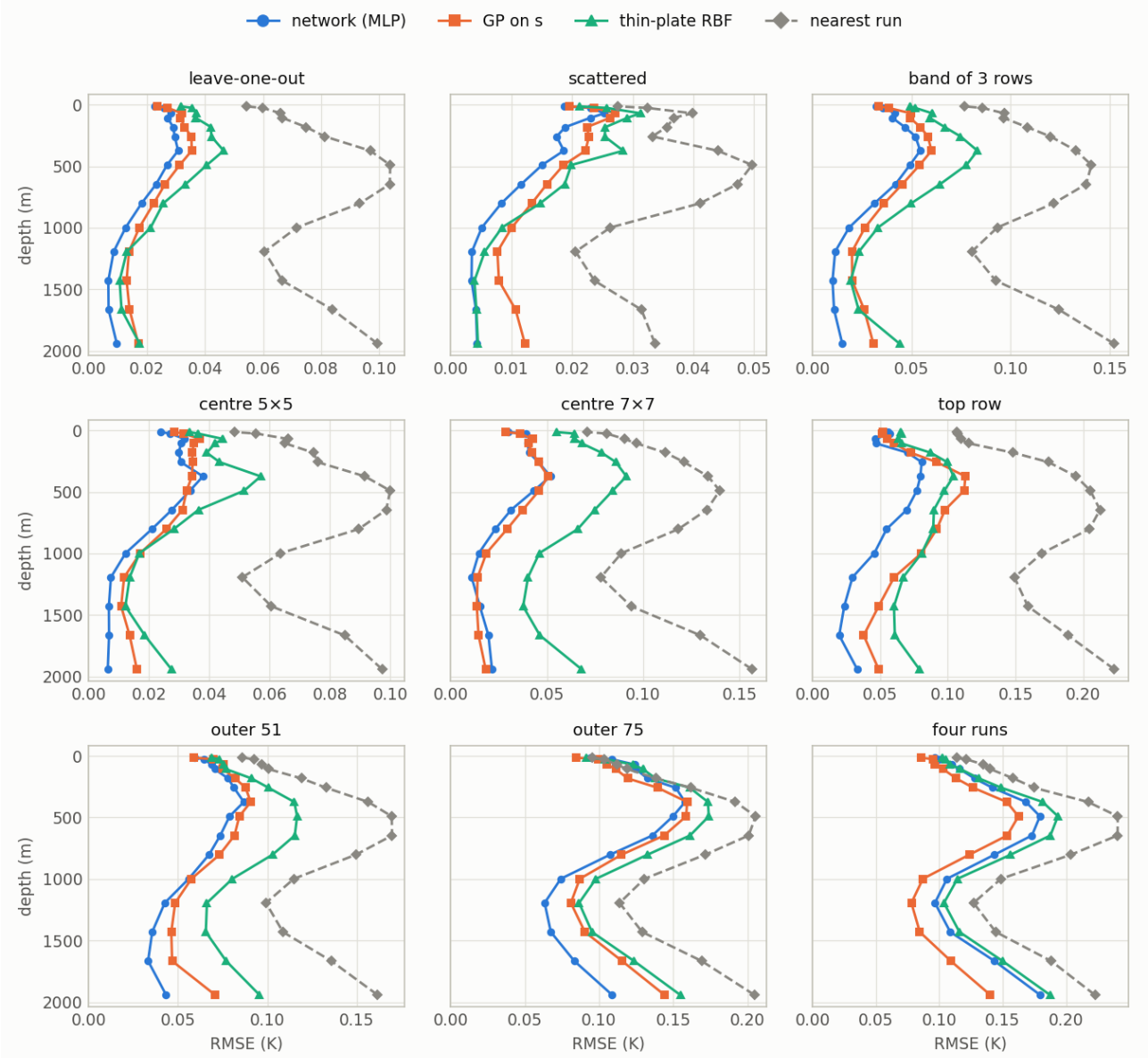


Figure 4: RMSE by depth, mean over the held-out runs of each split.

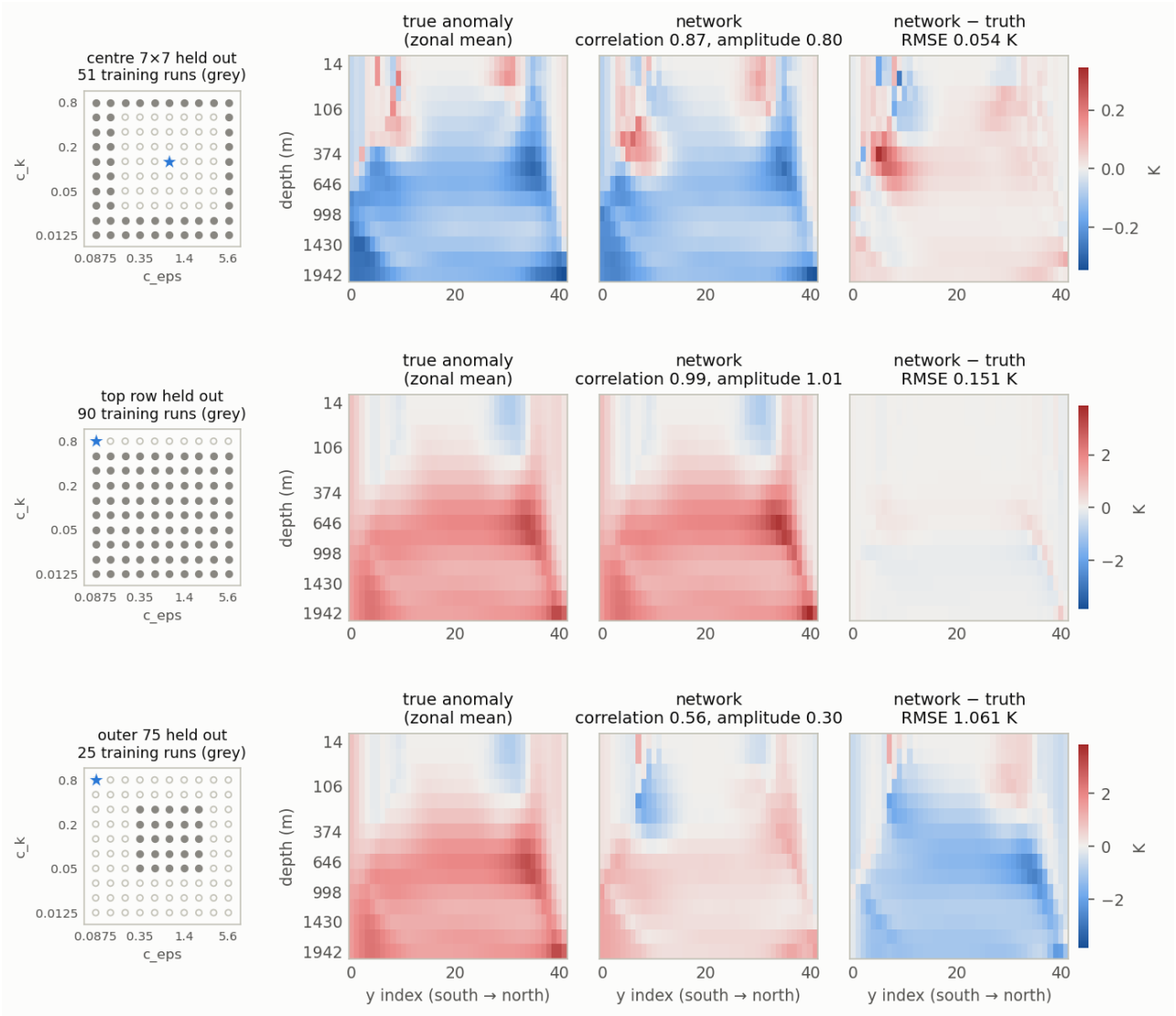


Figure 5: Zonal-mean sections of the last-year anomaly, relative to the mean of the 100 runs: truth, network and their difference, with one colour scale per run. Levels are drawn with equal height. Left: the split (grey: training runs; open circles: held out; star: the run shown). Top: the centre run of the 7×7 block, held out with the block. Middle and bottom: the warm-corner run, held out with the top row (90 training runs) and with the outer 75 (25 training runs). Correlation and amplitude (the regression slope of the predicted on the true anomaly) are over all water cells.